

Mincheol Son

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About

I am a security researcher specializing in cellular network security. My research focuses on vulnerabilities in LTE/5G/6G protocols, network systems, and baseband implementations. I design systematic testing and fuzzing frameworks to uncover security flaws and validate them through real-world networks and large-scale experiments.

Research Interests

- Cellular Network Security (LTE/5G/6G)
- Baseband Security
- Wireless System Security
- Fuzzing & Testing Frameworks
- LLM-Assisted Protocol Analysis & Security Testing

Education

Ph.D	KAIST (Korean Advanced Institute of Science and Technology) • Electrical Engineering • Advisor: Yongdae Kim	Sep. 2019 - Aug. 2026
MS	KAIST (Korean Advanced Institute of Science and Technology) • Electrical Engineering • Advisor: Yongdae Kim	Aug. 2017 - Sep. 2019
BS	Hanyang University , Electronic Engineering	Mar. 2013 - Aug. 2017

Publications

CITesting: Systematic Testing of Context Integrity Violations in LTE Core Networks Mincheol Son, Kwangmin Kim, Beomseok Oh, CheolJun Park, and Yongdae Kim <i>Distinguished Paper Award</i> <i>Proceedings of the 2025 on ACM SIGSAC Conference on Computer and Communications Security (ACM CCS 2025)</i>	2025 PDF 🔗
LTESniffer: An Open-source LTE Downlink/Uplink Eavesdropper Tuan Dinh Hoang, CheolJun Park, Mincheol Son, Taekkyung Oh, Sangwook Bae, Junho Ahn, Beomseok Oh, and Yongdae Kim <i>Proceedings of the 16th ACM Conference on Security and Privacy in Wireless and Mobile Networks (ACM WiSec 2023)</i>	2023 PDF 🔗
Preventing SIM Box Fraud Using Device Fingerprinting Beomseok Oh, Junho Ahn, Sangwook Bae, Mincheol Son, Yonghwa Lee, Min Suk Kang, and Yongdae Kim <i>Proceedings of the 30th Annual Network and Distributed Systems Security Symposium (NDSS 2023)</i>	2023 PDF 🔗
Watching the Watchers: Practical Video Identification Attack in LTE Networks Sangwook Bae, Mincheol Son, Dongkwan Kim, CheolJun Park, Jiho Lee, Sooel Son, and Yongdae Kim <i>Proceedings of the 31st USENIX Security Symposium (USENIX Security 2022)</i>	2022 PDF 🔗

Hiding in Plain Signal: Physical Signal Overshadowing Attack on LTE

Hojoon Yang, Sangwook Bae, **Mincheol Son**, Hongil Kim, Songmin Kim, and Yongdae Kim

Proceedings of the 28th USENIX Security Symposium (USENIX Security 2019)

2019
[PDF](#)

Hidden Figures: Comparative Latency Analysis of Cellular Networks with Fine-grained State Machine Models

Sangwook Bae, **Mincheol Son**, Sooel Son, and Yongdae Kim

Proceedings of the 20th International Workshop on Mobile Computing Systems and Applications (ACM HotMobile 2019)

2019
[PDF](#)

Awards

Distinguished Paper Award

2025

- ACM Conference on Computer and Communications Security (ACM CCS 2025)
- CITesting: Systematic Testing of Context Integrity Violations in LTE Core Networks

Best Paper Award

2025

- The 26th World Conference on Information Security Applications
- Denial-of-Service Risks in 5G Devices: Empirical Analysis on Connection Restriction Mechanisms

Cybersecurity Paper Competition 2025

2025

- Honorable Mention

Cybersecurity Paper Competition 2021

2021

- First Prize

Patents

KR 10-3016389 (Granted)

Sep. 2026

Method and Apparatus for Detecting Context Integrity Violations in Cellular Core Networks

KR 10-2025-0189113 (Filed)

Dec. 2025

Method and Apparatus for Memory Crash Detection Based on Network-Side State Control in Wireless Communication System

KR 10-3016382 (Granted)

Sep. 2026

Method and System for Generating and Managing Dynamic SUPI to Enhance User Security in 5G

KR 10-3016388 (Granted)

Sep. 2026

Dynamic Testing Method and System for Detecting Memory Corruptions in Lower Layers of Cellular Baseband of Mobile Communication Devices

KR 10-2997756 (Granted)

Jul. 2026

Malicious User Equipment-Based Security Analysis Method for Cellular Core Networks and the System Thereof

US 12,511,402 (Granted) / KR 10-2514797 (Granted)

Dec. 2025 / Mar. 2023

Security Analysis System and Method Based on Negative Testing for Protocol Implementation of LTE Device

KR 10-2678476 (Granted)

Jun. 2024

Method and Apparatus for Preventing SIM Box Fraud in Mobile Communication Network Using Device Fingerprinting

US 12,010,650 (Granted) / KR 10-2450114 (Granted)

Jun. 2024 / Sep. 2022

FBS Redirection Attack Method Using Unicast Message Injection in LTE and the System Thereof

KR 10-2546948 (Granted) Analysis System for Detection of SIP in VoLTE and the Method Thereof	Jun. 2023
KR 10-2514809 (Granted) Video Identification Method in LTE Networks and the System Thereof	Mar. 2023
US 11,405,787 (Granted) / KR 10-2515232 (Granted) Physical Signal Overshadowing Attack Method for LTE Broadcast Message and the System Thereof	Aug. 2022 / Mar. 2023
US 11,082,866 (Granted) / KR 10-2162024 (Granted) Apparatus and Method for Checking Problem of Mobile Communication Network Using Operational Logic Modeling and Comparative Analysis	Aug. 2021 / Sep. 2020

Projects

Development of Security Threat Response and Privacy Protection Technologies and International Cooperation for Establishing a Trusted Global Ecosystem in 6G NTN , Institute for Information & Communications Technology Planning & Evaluation (IITP)	2025 - 2027
Study on Network Security Inspection and Threat Assessment Against Subscriber Information-Based Attacks , SK Telecom Co., Ltd.	2025
Development of Private 5G Security Technology for Integrated Private 5G and Enterprise Network Security , Institute for Information & Communications Technology Planning & Evaluation (IITP)	2024 - 2027
Development of Anti-sniffing Technology in Mobile Communication and AirGap Environments , Institute for Information & Communications Technology Planning & Evaluation (IITP)	2024 - 2026
Automated Security Diagnosis Framework for Cellular Network Protocol using Formal and Comparative Analysis , Institute for Information & Communications Technology Planning & Evaluation (IITP)	2020 - 2023
Security Testing and Analysis of 5G Non-standalone Networks , LG Uplus Corp.	2020 - 2021
Intelligent 5G Core Network Abnormal Attack Detection & Countermeasure Technology Development , Institute for Information & Communications Technology Planning & Evaluation (IITP)	2019 - 2022
A Study on Physical Layer Security for Heterogeneous Wireless Network , Institute for Information & Communications Technology Planning & Evaluation (IITP)	2018 - 2022

Professional Activities

Secondary Reviewer

IEEE Symposium on Security and Privacy	2021 - 2022, 2024, 2026
USENIX Security Symposium (Security)	2020 - 2026
ACM Conference on Computer and Communications Security (CCS)	2021, 2023, 2025
Network and Distributed System Security Symposium (NDSS)	2020 - 2023, 2025
ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)	2023 - 2025
ACM International Conference on Mobile Computing and Networking (MobiCom)	2021
IEEE European Symposium on Security and Privacy (EuroS&P)	2020